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6

Introduction to Software

1. Which generation of programming languages is characterized by machine language?
 - a) First generation
 - b) Second generation
 - c) Third generation
 - d) Fourth generation
2. Which generation of programming languages introduced assembly language?
 - a) First generation
 - b) Second generation
 - c) Third generation
 - d) Fourth generation
3. Which generation of programming languages introduced high-level languages?
 - a) First generation
 - b) Second generation
 - c) Third generation
 - d) Fourth generation
4. Which generation of programming languages is associated with procedural programming?
 - a) First generation
 - b) Second generation
 - c) Third generation
5. Which generation of programming languages introduced object-oriented programming?
 - a) First generation
 - b) Second generation
 - c) Third generation
 - d) Fourth generation
6. Which generation of programming languages focuses on problem-solving through visual or graphical interfaces?
 - a) First generation
 - b) Second generation
 - c) Third generation
 - d) Fourth generation
7. Which generation of programming languages emphasizes on database and report generation?
 - a) First generation
 - b) Second generation
 - c) Third generation
 - d) Fourth generation
8. Which generation of programming languages is associated with natural language processing and AI capabilities?
 - a) First generation
 - b) Second generation

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- c) Third generation
- d) Fourth generation

9. Which generation of programming languages introduced declarative and functional programming?

- a) First generation
- b) Second generation
- c) Third generation
- d) Fourth generation

10. Which generation of programming languages is characterized by rapid application development and code reusability?

- a) First generation
- b) Second generation
- c) Third generation
- d) Fourth generation

11. What is the primary characteristic of First Generation Language (1GL)?

- a) It uses natural language for programming.
- b) It is based on binary codes that can be directly executed by the processor.
- c) It focuses on high-level abstractions and features.
- d) It eliminates the need for machine characteristics.

12. Which part of a machine language instruction conveys the

function to be performed by the instruction?

- a) Operation code
- b) Mode
- c) Operand
- d) Addressing type

13. What is the biggest limitation of using machine language for programming?

- a) It is difficult to understand and remember various combinations of 1's and 0's.
- b) It lacks support for basic arithmetic operations.
- c) It requires a deep understanding of machine hardware.
- d) It is not recognized by computer hardware.

14. Why is machine language considered error-prone?

- a) It requires extensive documentation for programming.
- b) It lacks support for complex programming tasks.
- c) It is challenging to locate and fix errors in machine instructions.
- d) It does not provide efficient memory management.

15. What is the major drawback of machine language in terms of portability?

- a) It requires modification when changing computer systems.

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- b) It is not compatible with modern computer architectures.
- c) It relies on external libraries for execution.
- d) It lacks support for input/output operations.

16. What makes machine language difficult to use and understand?

- a) The need to remember machine characteristics while programming.
- b) The lack of efficient memory management techniques.
- c) The absence of basic arithmetic operations.
- d) The limited support for input/output operations.

17. What is the primary advantage of machine language?

- a) It provides high-level abstractions for complex programming tasks.
- b) It enables efficient memory management.
- c) It is the native language of the computer and can be executed directly.
- d) It ensures portability across different computer systems.

18. What happens when a computer system is changed while using machine language programs?

- a) The programs automatically adapt to the new system.

- b) The programs become obsolete and need to be rewritten in the new machine code.
- c) The programs require minimal modifications for compatibility.
- d) The programs continue to run without any changes.

19. Which component(s) of a computer system affects the variation of machine language across different computer types?

- a) ALU (Arithmetic Logic Unit)
- b) CU (Control Unit)
- c) Memory unit
- d) All of the above

20. What is the native language of the computer that the CPU can directly execute?

- a) Assembly language
- b) High-level language
- c) Machine language
- d) Scripting language

21. What is the primary characteristic of Second Generation Language (2GL)?

- a) It uses natural language for programming.
- b) It is based on binary codes that can be directly executed by the processor.
- c) It uses mnemonic codes to represent instructions.
- d) It focuses on high-level abstractions and features.



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22. What does an assembly language statement typically consist of?

- a) Label, operation code, and operands
- b) Label, binary code, and operands
- c) Operation code, binary code, and operands
- d) Operation code, label, and operands

23. What is the purpose of a label in assembly language?

- a) It represents the register where data is located.
- b) It specifies the particular operation to be performed.
- c) It identifies and references instructions in the program.
- d) It represents the binary code for an instruction.

24. How is an assembly language program translated into machine language?

- a) By a compiler
- b) By an interpreter
- c) By an assembler
- d) By a linker

25. Why is assembly language referred to as a low-level language?

- a) It has limited support for high-level abstractions.

b) It requires complex memory management techniques.

c) It is closer to the binary machine language of the computer.

d) It is primarily used in embedded systems programming.

26. What is the main advantage of writing a program in assembly language compared to machine language?

- a) Improved readability due to symbolic instructions.
- b) Better support for complex arithmetic operations.
- c) Higher level of portability across different computer systems.
- d) Easier debugging and error correction.

27. What is the major disadvantage of assembly language?

- a) It requires a deep understanding of the internal architecture of the processor.
- b) It lacks support for basic arithmetic operations.
- c) It is difficult to read and understand due to binary sequences.
- d) It is not recognized by computer hardware.

28. Why is assembly language considered machine dependent?

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- a) It relies on external libraries for execution.
- b) It lacks support for input/output operations.
- c) It is specific to the internal architecture of a particular processor model.
- d) It requires constant updates and modifications.

29. How does programming with assembly language compare to machine language in terms of difficulty and time consumption?

- a) It is easier and less time-consuming than machine language.
- b) It is more difficult and time-consuming than machine language.
- c) It has similar difficulty and time consumption as machine language.
- d) It depends on the specific program and task at hand.

30. What is the role of an assembler in assembly language programming?

- a) It executes the assembly language program directly.
- b) It translates assembly language code into machine language.
- c) It provides a user-friendly interface for writing assembly programs.
- d) It manages memory allocation and deallocation.

31. What happens if an assembly language program written for one processor is executed on a different processor?

- a) It runs without any issues due to the compatibility of assembly language.
- b) It requires minor modifications to make it compatible with the new processor.
- c) It produces errors or does not work correctly due to architectural differences.
- d) It requires recompilation using a different assembly language.

32. Which of the following statements about assembly language is true?

- a) It is widely used for artificial intelligence research.
- b) It is a high-level language with advanced features.
- c) It is primarily used in web development.
- d) It provides direct access to hardware and low-level programming control.

33. What does the term "mnemonic code" refer to in assembly language?

- a) The binary representation of an instruction.
- b) The label assigned to an instruction.

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- c) The operation code indicating the function to be performed.
d) The set of instructions for mathematical calculations.
34. What is the level of abstraction in assembly language compared to high-level languages?
a) Higher level of abstraction
b) Lower level of abstraction
c) Similar level of abstraction
d) No abstraction is provided
35. How does assembly language programming differ from writing programs in high-level languages?
a) Assembly language provides better code optimization.
b) High-level languages offer improved readability and ease of use.
c) Assembly language programs are executed faster than high-level language programs.
d) High-level languages are machine-dependent like assembly language.
36. What type of operations are commonly performed in assembly language?
a) Logical operations
b) Database operations
c) Networking operations
d) Arithmetic and data manipulation operations
37. What is the purpose of an operand in assembly language?
a) It represents the label assigned to an instruction.
b) It specifies the operation code for an instruction.
c) It indicates the function to be performed by an instruction.
d) It represents the data or memory location involved in an operation.
38. What is the primary reason for using assembly language instead of machine language?
a) Assembly language offers higher-level abstractions.
b) Assembly language programs are more portable across different systems.
c) Assembly language provides better error detection and correction.
d) Assembly language code is easier to read and understand.
39. Which of the following is an example of an assembly language instruction?
a) MOV 5, R1
b) IF-ELSE statement
c) FOR loop
d) FUNCTION declaration
40. Which of the following components translates assembly language code into machine language?

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- a) Compiler
- b) Interpreter
- c) Assembler
- d) Linker

41. Which generation of programming languages is commonly referred to as high-level languages?

- a) First generation language (1GL)
- b) Second generation language (2GL)
- c) Third generation language (3GL)
- d) Fourth generation language (4GL)

42. What was the primary goal behind the development of high-level languages?

- a) To make programming more efficient
- b) To provide low-level control over the hardware
- c) To make programming easier and less error-prone
- d) To ensure compatibility across different processors

43. Which of the following is NOT an example of a high-level programming language?

- a) C
- b) COBOL
- c) Assembly language
- d) Python

44. Which programming languages have instructions that are similar to the English language?

- a) First generation languages
- b) Second generation languages
- c) Third generation languages
- d) Fourth generation languages

45. What is the nature of most third generation languages?

- a) Procedural
- b) Object-oriented
- c) Functional
- d) Declarative

46. What must be done before a program written in a high-level language can be executed by a computer?

- a) The program must be converted to assembly language.
- b) The program must be converted to machine language.
- c) The program must be interpreted by a compiler.
- d) The program must be optimized for performance.

47. Which component is responsible for translating a high-level language program into machine language?

- a) Compiler
- b) Interpreter
- c) Assembler
- d) Debugger

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48. How does the readability of programs written in high-level languages compare to those written in assembly and machine languages?

- a) Programs in high-level languages are less readable.
- b) Programs in high-level languages are equally readable.
- c) Programs in high-level languages are more readable.
- d) Readability depends on the specific language used.

49. Which advantage of high-level programming languages allows for easy detection and removal of errors?

- a) Readability
- b) Portability
- c) Easy debugging
- d) Ease in software development

50. What motivated the development of high-level languages?

- a) Low-level control over the hardware
- b) Compatibility across different processors
- c) High development costs of machine and assembly language programs
- d) Need for faster program execution

51. What is the primary characteristic of Fourth Generation Languages (4GL)?

- a) They are machine-dependent.
- b) They are procedural in nature.
- c) They are primarily used for system-level programming.
- d) They provide a higher level of abstraction and require less coding.

52. Which of the following is an example of a Fourth Generation Language (4GL)?

- a) C
- b) Assembly language
- c) COBOL
- d) SQL

53. What is the main advantage of using Fourth Generation Languages (4GL)?

- a) They provide low-level control over the hardware.
- b) They offer improved portability across different systems.
- c) They require extensive knowledge of computer architecture.
- d) They allow for faster development and higher productivity.

54. What type of tasks are Fourth Generation languages (4GL) commonly used for?

- a) Scientific calculations and mathematical modeling.

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b) System administration and hardware configuration.

c) Writing operating systems and device drivers.

d) Database querying and report generation.

55. Which statement accurately describes the syntax of Fourth Generation Languages (4GL)?

a) They use English-like statements.

b) They use binary code for programming.

c) They rely on mnemonic codes for instructions.

d) They require explicit memory management.

56. Which of the following is NOT a characteristic of Fourth Generation Languages (4GL)?

a) They are closer to natural language.

b) They provide high-level abstractions.

c) They require extensive hardware knowledge.

d) They focus on data manipulation and database operations.

57. Which component is responsible for translating Fourth Generation Language (4GL) code into machine-executable instructions?

a) Compiler

b) Interpreter

c) Assembler

d) Debugger

58. What is the primary focus of Fourth Generation Languages (4GL)?

a) Low-level hardware control.

b) Efficient memory management.

c) Simplified and rapid application development.

d) Real-time system programming.

59. Which of the following tasks is commonly performed using Fourth Generation Languages (4GL)?

a) Operating system kernel development.

b) Device driver programming.

c) Business process automation.

d) Network protocol design.

60. What is the level of abstraction offered by Fourth Generation Languages (4GL)?

a) Low-level, close to the hardware.

b) Intermediate, between assembly and high-level languages.

c) High-level, providing a more user-friendly approach.

d) System-level, focusing on low-level system interactions.

61. What is the primary characteristic of Fifth Generation Languages (5GL)?

a) They are machine-dependent.

b) They are procedural in nature.



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- c) They are primarily used for system-level programming.
- d) They are based on solving problems using constraints.

62. Which of the following is an example of a Fifth Generation Language (5GL)?

- a) C
- b) Java
- c) FORTRAN
- d) Prolog

63. What is the main focus of Fifth Generation Languages (5GL)?

- a) Low-level hardware control.
- b) Database management and querying.
- c) Constraint-based problem solving.
- d) Efficient memory management.

64. What distinguishes Fifth Generation Languages (5GL) from lower-level languages?

- a) They use assembly language instructions.
- b) They require extensive hardware knowledge.
- c) They focus on artificial intelligence research.
- d) They are based on constraints rather than algorithms.

65. Which of the following tasks are Fifth Generation Languages (5GL) commonly used for?

- a) Writing operating systems.
- b) Scientific calculations and mathematical modeling.
- c) Database administration and management.
- d) Solving complex problems in artificial intelligence.

66. Which statement accurately describes the syntax of Fifth Generation Languages (5GL)?

- a) They use English-like statements.
- b) They use binary code for programming.
- c) They rely on mnemonic codes for instructions.
- d) They require explicit memory management.

67. Which component is responsible for translating Fifth Generation Language (5GL) code into machine-executable instructions?

- a) Compiler
- b) Interpreter
- c) Assembler
- d) Debugger

68. What is the primary focus of Fifth Generation Languages (5GL)?

- a) Low-level hardware control.
- b) Efficient memory management.

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- c) Solving complex problems using constraints.
- d) Real-time system programming.

69. Which of the following tasks is commonly performed using Fifth Generation Languages (5GL)?

- a) Writing device drivers.
- b) Network protocol design.
- c) Natural language processing.
- d) Business process automation.

70. What is the level of abstraction offered by Fifth Generation Languages (5GL)?

- a) Low-level, close to the hardware.
- b) Intermediate, between assembly and high-level languages
- c) High-level, providing a more user-friendly approach.
- d) System-level, focusing on low-level system interactions.

71. Which of the following is not a commonly used classification of programming languages?

- a) High-level languages vs. low-level languages
- b) Procedural languages vs. object-oriented languages
- c) Compiled languages vs. interpreted languages
- d) Sequential languages vs. parallel languages

72. High-level languages are characterized by:

- a) Being closely tied to the hardware architecture
- b) Being easy to understand and write for humans
- c) Directly executing machine code without translation
- d) Being primarily used for system-level programming

73. Which classification is based on how instructions are organized and executed in a program?

- a) Procedural languages vs. object-oriented languages
- b) Sequential languages vs. parallel languages
- c) Compiled languages vs. interpreted languages
- d) Imperative languages vs. declarative languages

74. Which type of language requires the use of a compiler to translate the code into machine-executable instructions?

- a) Interpreted languages
- b) High-level languages
- c) Assembly languages
- d) Compiled languages

75. Which classification distinguishes languages based on their approach to managing memory and system resources?

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- a) Procedural languages vs. object-oriented languages
- b) Sequential languages vs. parallel languages
- c) High-level languages vs. low-level languages
- d) Compiled languages vs. interpreted languages

76. Object-oriented languages are characterized by:

- a) Having a linear execution flow
- b) Emphasizing the use of functions and procedures
- c) Encapsulating data and behavior into objects
- d) Being primarily used for scientific calculations

77. Which classification is based on the style or paradigm of programming employed by the language?

- a) Procedural languages vs. object-oriented languages
- b) Compiled languages vs. interpreted languages
- c) Imperative languages vs. declarative languages
- d) High-level languages vs. low-level languages

78. Declarative languages focus on:

- a) Executing instructions in a specific order

- b) Defining the steps and procedures to solve a problem
- c) Specifying what needs to be achieved, rather than how
- d) Manipulating objects and their properties

79. Which classification considers the level of abstraction provided by the language?

- a) Sequential languages vs. parallel languages
- b) High-level languages vs. low-level languages
- c) Compiled languages vs. interpreted languages
- d) Procedural languages vs. object-oriented languages

80. Which type of language allows for concurrent execution of multiple tasks or processes?

- a) Procedural languages
- b) Assembly languages
- c) Parallel languages
- d) Interpreted languages

81. What is the purpose of compiling a program?

- a) To translate the program into machine code
- b) To check for syntax errors in the program
- c) To optimize the program for better performance
- d) To combine multiple source code files into a single executable

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82. Which stage of the software development process involves converting the source code into machine code?

- a) Compiling
- b) Linking
- c) Loading
- d) Debugging

83. What does the linker do during the linking stage?

- a) Combines object code files and libraries to create an executable file
- b) Translates the source code into machine code
- c) Checks the program for logical errors
- d) Optimizes the program for better performance

84. Which of the following is true about loading a program?

- a) It is performed before the compiling stage.
- b) It converts the source code into machine code.
- c) It prepares the program for execution by allocating memory.
- d) It checks the program for syntax errors.

85. What is the purpose of the symbol table in the linking process?

- a) It contains the names and memory addresses of variables and functions.
- b) It checks for syntax errors in the program.
- c) It optimizes the program for better performance.
- d) It combines multiple source code files into a single executable.

86. During the linking stage, unresolved symbols are resolved by:

- a) The compiler
- b) The linker
- c) The loader
- d) The debugger

87. Which of the following is responsible for allocating memory for a program's executable code, data, and libraries?

- a) Compiler
- b) Linker
- c) Loader
- d) Debugger

88. Which stage of the software development process is responsible for identifying and resolving external references between different object code files?

- a) Compiling
- b) Linking
- c) Loading
- d) Debugging

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89. Which of the following is true about dynamic linking?

- a) It is performed during the compiling stage.
- b) It combines multiple source code files into a single executable.
- c) It allows multiple programs to share the same copy of a library in memory.
- d) It converts the source code into machine code.

90. What is the role of the loader in the software execution process?

- a) It translates the source code into machine code.
- b) It checks for syntax errors in the program.
- c) It prepares the program for execution by allocating memory and resolving external references.
- d) It combines multiple source code files into a single executable.

91. Which of the following is responsible for translating the entire source code into machine code?

- a) Compiler
- b) Assembler
- c) Interpreter
- d) Linker

92. Which of the following is responsible for converting assembly language code into machine code?

- a) Compiler

b) Assembler

- c) Interpreter
- d) Loader

93. The process of translating high-level source code into machine code is known as:

- a) Compilation
- b) Assembly
- c) Interpretation
- d) Loading

94. What is the primary purpose of a loader?

- a) To translate the source code into machine code
- b) To check for syntax errors in the program
- c) To allocate memory and prepare the program for execution
- d) To combine multiple object code files into a single executable

95. Which component resolves external references between different object code files?

- a) Compiler
- b) Assembler
- c) Interpreter
- d) Linker

96. Which of the following is responsible for allocating memory for a program's code and data during the loading process?

- a) Compiler
- b) Assembler

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- c) Loader
- d) Linker

97. The process of combining multiple object code files and libraries to create a single executable file is performed by the:

- a) Compiler
- b) Assembler
- c) Loader
- d) Linker

98. Which component performs the translation of the source code into machine code line by line as it is executed?

- a) Compiler
- b) Assembler
- c) Interpreter
- d) Linker

99. Which of the following is true about interpreters?

- a) They generate an executable file that can be directly executed by the processor.
- b) They produce machine code that is independent of the target hardware.
- c) They translate the entire source code into machine code before execution.
- d) They execute the source code directly, without generating intermediate machine code.

100. Which component is responsible for resolving symbols

and addresses in the final executable file?

- a) Compiler
- b) Assembler
- c) Interpreter
- d) Linker

101. Which system software is responsible for translating a high-level source program into a form the machine can execute?

- a) Compiler
- b) Interpreter
- c) Assembler
- d) Translator

102. What is the purpose of the linker in the software development process?

- a) Translating high-level source code into machine code
- b) Executing the program line by line
- c) Resolving symbolic references between object programs
- d) Allocating memory and preparing the program for execution

103. What does relocation refer to in the context of programming?

- a) Translating high-level code into machine code
- b) Combining multiple object programs into a single executable
- c) Assigning addresses to various parts of the program
- d) Resolving symbolic references between object programs



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104. Which system software places an executable program's instructions and data from secondary memory into primary memory and prepares them for execution?

- a) Compiler
- b) Interpreter
- c) Assembler
- d) Loader

105. Which process adjusts the code and data in a program to reflect the assigned addresses?

- a) Translation
- b) Linking
- c) Relocation
- d) Loading

106. The term "linking" in the software development process refers to:

- a) Translating high-level code into machine code
- b) Combining multiple source files into a single executable
- c) Resolving symbolic references between object programs
- d) Allocating memory and preparing the program for execution

107. What is the purpose of a loader in the software execution process?

- a) Translating high-level code into machine code
- b) Executing the program line by line

- c) Resolving symbolic references between object programs
- d) Placing executable program instructions and data into primary memory

108. Which system software assigns addresses to various parts of the program during the compilation process?

- a) Compiler
- b) Interpreter
- c) Assembler
- d) Linker

109. What is the main goal of relocation in program execution?

- a) Combining multiple object programs into a single executable
- b) Resolving symbolic references between object programs
- c) Allocating memory and preparing the program for execution
- d) Adjusting the code and data in the program to reflect assigned addresses

110. The purpose of the translator is to:

- a) Translate high-level code into machine code
- b) Combine multiple source files into a single executable
- c) Resolve symbolic references between object programs
- d) Execute the program line by line



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Explanation

1. The correct answer is a) First generation. The first generation of programming languages was characterized by machine language, which consists of instructions written in binary code that can be directly executed by the computer's hardware.

2. The correct answer is b) Second generation. The second generation of programming languages introduced assembly language. Assembly language uses mnemonic codes to represent machine instructions, making it easier for programmers to write and understand instructions compared to machine language.

3. The correct answer is c) Third generation. The third generation of programming languages introduced high-level languages. High-level languages are designed to be closer to human language and provide higher-level abstractions and features, making programming more efficient and accessible.

4. The correct answer is c) Third generation. The third generation of programming languages, such as Fortran, COBOL, and C, is associated with procedural programming. Procedural programming focuses on sequences of instructions and subroutines for solving problems.

5. The correct answer is c) Third generation. The third generation of programming languages, such as C++ and Java, introduced object-oriented programming (OOP). OOP allows for the creation of objects that encapsulate data and behavior, promoting modularity, reusability, and better software design.

6. The correct answer is d) Fourth generation. The fourth generation of programming languages focuses on problem-solving through visual or graphical interfaces. It emphasizes on rapid application development and enables users to interact with the software through intuitive visual tools.

7. The correct answer is d) Fourth generation. The fourth generation of programming languages



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emphasizes on database and report generation. These languages provide powerful features for handling data and generating reports from databases.

8. The correct answer is d) Fourth generation. The fourth generation of programming languages introduced advanced capabilities such as natural language processing and AI. These languages enable the development of intelligent systems that can understand and process human-like language.

9. The correct answer is c) Third generation. While the third generation of programming languages introduced high-level languages, it also paved the way for declarative and functional programming paradigms. Functional languages, such as Lisp and Haskell, focus on computations based on mathematical functions.

10. The correct answer is d) Fourth generation. The fourth generation of programming languages is characterized by rapid application development and code reusability. These languages provide frameworks, libraries, and tools that facilitate faster development cycles

and the reuse of existing code components.

Remember, the generations of programming languages represent different stages in the development and evolution of programming languages, each introducing new features and capabilities to enhance the programming experience and enable more efficient and powerful software development.

11. The correct answer is b) It is based on binary codes that can be directly executed by the processor. First Generation Language (1GL) or machine language instructions are written in binary codes that can be immediately executed by the processor. It is the most basic form of programming language understood by the computer hardware.

12. The correct answer is a) Operation code. The operation code is the part of a machine language instruction that conveys to the computer what function or operation needs to be performed. It specifies the desired operation, such as addition, subtraction, or moving data.

13. The correct answer is a) It is difficult to understand and

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remember various combinations of 1's and 0's. Programming in machine language can be challenging due to the need to remember and work with the binary representation of data and instructions. The programmer must have a deep understanding of the machine characteristics and the specific binary codes used for different operations.

14. The correct answer is c) It is challenging to locate and fix errors in machine instructions. Since machine language instructions are written in binary codes, locating errors in the instructions can be a tedious and error-prone process. The programmer must carefully examine the binary codes to identify any mistakes or inconsistencies.

15. The correct answer is a) It requires modification when changing computer systems. Machine language is closely tied to the specific hardware architecture of a computer system. If the computer system is changed, the programmer may need to learn a new machine code and rewrite all the existing programs to be compatible with the new system.

16. The correct answer is a) The need to remember machine characteristics while programming. Machine language programming requires the programmer to have a deep understanding of the specific machine characteristics, including the binary codes used for different operations. This makes it difficult to develop and understand programs in machine language.

17. The correct answer is c) It is the native language of the computer and can be executed directly. One advantage of machine language is that it is the native language of the computer. It can be executed directly by the CPU without the need for translation or interpretation, resulting in efficient and fast execution.

18. The correct answer is b) The programs become obsolete and need to be rewritten in the new machine code. When a computer system is changed, the existing machine language programs become incompatible with the new system. The programmer would need to learn the new machine code and rewrite all the existing programs to make them work on the new system.

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19. The correct answer is d) All of the above. The variation of machine language across different computer types is influenced by various components of a computer system, including the ALU (Arithmetic Logic Unit), CU (Control Unit), and the memory unit. Different computer architectures have different designs and requirements, leading to variations in machine language.

20. The correct answer is c) Machine language. Machine language is the native language of the computer, consisting of binary codes that the CPU can directly execute. It is the lowest-level programming language understood by the computer hardware.

Remember, First Generation Language (1GL) or machine language is the most basic form of programming language, written in binary codes that can be executed directly by the processor. It requires a deep understanding of machine characteristics and has limitations in terms of complexity, portability, and ease of use compared to higher-level programming languages.

21. The correct answer is c) It uses mnemonic codes to represent instructions. Assembly language uses mnemonic codes, which are

symbolic representations of instructions, making it easier for programmers to write and understand the code compared to binary machine language.

22. The correct answer is a) Label, operation code, and operands. An assembly language statement typically consists of a label (optional), an operation code (mnemonic), and one or more operands. Labels are used for identification and referencing, the operation code specifies the operation to be performed, and operands represent the data or memory locations involved.

23. The correct answer is c) It identifies and references instructions in the program. Labels in assembly language are used to identify and reference instructions within the program. They provide a way to jump or branch to specific locations in the code.

24. The correct answer is c) By an assembler. An assembler is a translator that converts assembly language code into machine language code. It produces a single machine language instruction for each assembly language statement,